

Helion: the next diagnostic decision

Which procedure should the engineer run next?

Lower investigation cost and turnaround, subject to the same diagnostic completeness and safety requirements.

1. Does manufacturing context improve **fault prediction**?
2. Do predictions improve **diagnostic selection** beyond rules?
3. Can dispatch improve the existing **scheduling examples**?

The engineer owns diagnosis and closure. A proposed shift coordinator owns dispatch. Coexisting faults remain separate diagnostic obligations.

Source: Client brief; expanded user scope; research_v2 concern-closure benchmark.

Presenter notes

Diagnostic test selection

Time: 0:00–0:45

Helion wants the on-duty engineer to choose the next diagnostic procedure while accounting for cost and useful completeness. Our research tests fault prediction and the next-test choices it supports. Better prediction alone need not change the work required. A real coexisting fault remains useful even when it adds work. The engineer retains procedure and closure authority. The headline finding is that the model predicts some synthetic faults usefully, but adds no demonstrated diagnostic value over the candidate rules in this experiment.

Evidence: Client brief; design notebook, Helion context; benchmark report.

The experiment and its comparators

Offline pipeline: validate, train, score, replay, schedule and report.

Policy	Next-procedure choice
Candidate rules	Open concerns and co-fault safeguards in priority order
CT-first compatibility	Same v2 priority; retained as a regression comparator
Four heuristics	Coverage/cost with prevalence or one of three models

Shared standard: routine concerns and failed branches resolved; audits still require all-seven evidence.

One conditional repeat follows an inconclusive nondestructive attempt. Unfinished work stays pending; a low score cannot establish absence. Unresolved intact-sample work blocks destructive SEM.

Source: Pipeline README; MOCK-ENG-002; benchmark §2. Same concerns, costs, scope and paired report draws.

Presenter notes

Implemented pipeline and fair comparators

Time: 0:45–1:45

Acceptance has already rejected the stack. Diagnostic findings inform later process investigation without proving which machine caused a defect. We implemented six policies with the same costs, concern-opening rules, co-fault safeguards and report scope. The candidate rules and CT-first compatibility arm use the same v2 priority; heuristic arms may reorder eligible concern work. Findings change evidence and eligibility while initial probabilities remain fixed. A low probability never confirms absence. A gross-delamination CT negative cannot exclude all delamination, and IR provides localisation only. Open intact-sample obligations or contradictions block destructive SEM. These restrictions let us compare policies without rewarding omitted required work.

Evidence: Pipeline README; MOCK-ENG-002; benchmark report, section 2.

Data and operating assumptions

Supplied synthetic data

916 rejected stacks, seven fault labels

653 / 126 / 137

training / validation / test

Test: **6** die-crack cases and
2 cases with coexisting faults

Resource dollars per attempt: CT \$120 acoustic \$200 electrical \$600 IR \$150 SEM \$1,800

Some allowed acceptance readings directly encode generated faults. The already-inspected test cohort provides retrospective evidence.

Still needed: real procedure histories, qualified SOP, complete audit findings and observed costs.

Source: Validation record; benchmark evidence boundary; SYN-OPS-001. Staff and costs are invented assumptions.

Assumed operations

8 technicians and 8 diagnostic engineers

Skills, one-day calendars and reservations

Dollar rates, task phases and procedure
error rates

Presenter notes

Evidence and operating assumptions

Time: 1:45–3:00

We retained the 653, 126 and 137 rejected-stack partitions with lot/time separation. All training targets come from supplied synthetic truth. Preprocessing fits training data only. IDs, later annotations, simulator internals and timing margin are excluded. Even so, permitted acceptance readings contain shortcuts: the detected-interconnect count equals the generated TSV-plus-microbump fault count. The manufacturing-only comparison exposes dependence on those inputs. Test data was already inspected, so findings are retrospective. Six crack cases and two coexisting cases limit conclusions. Dollar rates and report-error rates are assumptions. Costs charge technician and engineer labour, equipment occupancy and supplies. They represent consumed resources, not wholly avoidable cash. Forty cases received an independent synthetic audit assignment, fixed across policies. Real independent complete audits remain missing.

Evidence: validation.json; SYN-OPS-001; benchmark evidence boundary.

Fault prediction on the test cohort

137 retrospective test cases. Seven binary logistic heads allow coexisting faults.

Input comparison	Macro AP $\uparrow$	Log loss $\downarrow$
Constant training prevalence	0.145	0.402
Inspection/acceptance	0.504	0.231
Full allowed context	0.497	0.251
Manufacturing/package only	0.189	0.408

0.497 is macro average precision: a ranking summary averaged equally across seven faults. It does not mean 49.7% of diagnoses are correct.

Full-minus-inspection AP: -0.007 ; paired 95% interval $[-0.058, +0.047]$. The small AP difference is uncertain.

Source: predictive_metrics.csv and predictive_paired_comparisons.csv, test rows. Synthetic shortcuts limit generalisation.

Presenter notes

Fault-prediction findings

Time: 3:00–4:30

We fitted regularised logistic regression with training-only imputation, missingness indicators, scaling and categorical encoding. All learned alternatives used the same training-lot folds and three-value regularisation search. A single saved artifact retains the research alternatives. Each has seven binary heads, so faults may coexist. Average precision summarises precision and recall across ranking thresholds. Macro averaging gives every mechanism equal weight. It differs from precision at a selected threshold and from diagnosis accuracy. Both acceptance-based variants exceed the 0.145 prevalence reference, but full context does not improve the point estimates. The paired 95% interval for full-minus-inspection macro AP spans -0.058 to $+0.047$, so the small AP difference is uncertain. Full-model Brier score is 0.081 versus inspection's 0.077, with lower preferred. Calibration was assessed without fitting an extra calibrator. Strong electrical prediction and weak rare-fault prediction should not disappear inside one average.

Evidence: predictive_metrics.csv; predictive_paired_comparisons.csv; training.json; calibration figure.

ML changes order, not the conclusion

Base test replay: 137 cases × 100 paired report replications per policy.

Policy	Consumed USD/case	Evidence complete	Correctly complete
Candidate rules	\$1,169.26	86.6%	82.3%
CT-first compatibility	\$1,169.26	86.6%	82.3%
Full-model heuristic	\$1,170.92	86.6%	82.3%

Candidate rules and CT-first are identical in v2. The full model changes about **5.9% of sequences**, with no material completion change.

Full minus candidate rules: **+\$1.66/case**; 95% lot-bootstrap interval **[+\$0.51, +\$2.98]**. Missed faults improve by about 0.002/case.

Every case counts, including unfinished investigations. Full model: \$60.44/case in known pending procedures. Unknown review continuation remains unpriced.

Source: decision_metrics.csv; decision_uncertainty.csv; verification.json. Synthetic resource costs, not fab savings.

Presenter notes

Diagnostic selection versus the baselines

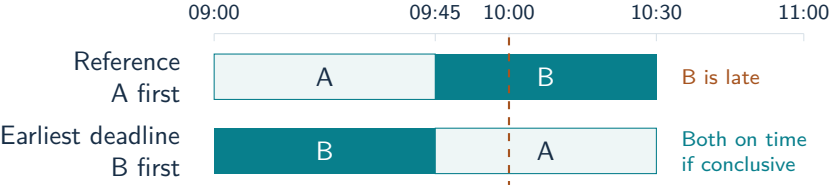
Time: 4:30–6:15

Policies received the same potential reports for each sample, procedure and attempt, with shared audit assignments. A hundred replications vary assumed report outcomes on the same cases; they do not create new independent samples. Candidate rules and CT-first are identical under the v2 concern-priority implementation. The full model changes about 5.9% of sequences, costs \$1.66 more per case and slightly lowers missed faults, while rounded completion and correctly-complete rates remain unchanged. That is not demonstrated operational value. Pending dollars cover known unattempted procedures only. Manual review and future resolution remain unpriced. Conditional nondestructive repeats and blocked destructive work keep inconclusive cases visible. The 1,000 paired lot-cluster bootstrap resamples quantify case uncertainty separately from Monte Carlo variation. Cost and report-error stresses test assumptions without establishing real-world benefit.

Evidence: decision_metrics.csv (base/test); decision_uncertainty.csv; verification.json; benchmark report.

Dispatch changes which milestone is met

Case D: resources unavailable before 09:00. B's warpage milestone is 10:00; A's underfill milestone is 11:00.



Both orders cost \$240 and finish at 10:30. Staff phases, equipment and specimen exclusivity remain feasible.

Inconclusive findings stay unresolved. Scheduling covers the existing 24-hour fixtures only; cohort hours are nominal procedure totals.

Source: scheduling.json; SYN-OPS-001 Case D. Scoped milestones depend on conclusive findings. Actual dispatch is unknown.

Presenter notes

Dispatch changes which milestone is met

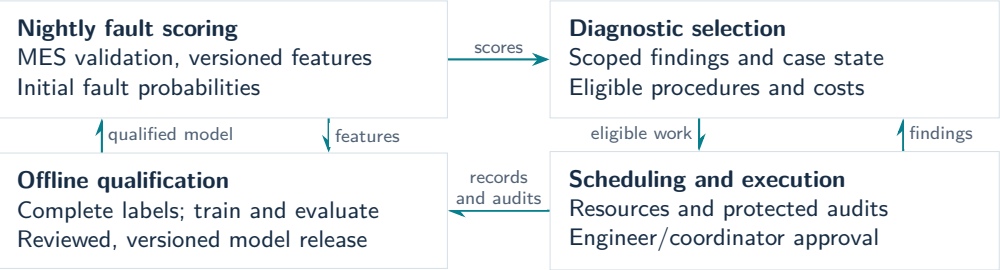
Time: 6:15–7:30

The scheduler uses the existing one-day snapshot, preserving commitments before 09:00. TECH-01 prepares each stack for 15 minutes and QE-01 interprets the final 15 minutes; those phases and equipment reservations do not conflict. A-first misses B's milestone even though both orders consume the same resources. This demonstrates dispatch feasibility for the documented cases, not a quarterly turnaround gain. An inconclusive B remains unresolved and does not cancel A's accepted booking. Case B has \$920 spent with SEM pending; Case C's \$2,720 conclusive path remains a conditional cost trace because the supplied calendar cannot support a future SEM appointment.

Evidence: scheduling.json; walkthroughs.json; SYN-OPS-001 Case D.

Proposed fab operation and feedback

Offline prototype implemented. Diagram: proposed fab integration and qualification.



Monitor: batch health, input drift, audited errors, queue age, inconclusives and overrides.

Fallback: rules/manual review for failed imports, unqualified contexts or stale calendars. Incomplete audits remain incomplete.

Source: Design blueprint and architecture; pipeline README. Yield Engineering owns outcomes, lab operations calendars, IT ingestion.

Presenter notes

Proposed operation, monitoring and fallback

Time: 7:30–9:00

The diagram remains the proposed fab architecture. Our implementation is an offline CLI with saved preprocessing, recorded seeds, dependencies and checksums. It installs no nightly job, interface or production service. Proposed nightly scoring fits the local environment, while new diagnostic findings would update eligibility between imports. Failed imports produce no fresh scores. Missing required records or unknown tool/supplier conditions route to rules or manual review. Monitoring would separate input drift from audited outcome deterioration. Unresolved investigations, inconclusive frequency and engineer overrides are early proxies while labels arrive. Yield Engineering would review model outcomes and IT would investigate ingestion failures. New tools or suppliers trigger qualification review. Independent audit selection remains separate from model recommendations. Only qualified, complete audit findings enter the reference-label cohort. Inconclusive or contradictory audits remain incomplete. Only reviewed candidates could replace a qualified model. These operational controls still require real records and prospective testing.

Evidence: Design notebook, operations; helion-architecture.mmd; pipeline README; run_manifest.json.

What the evidence supports next

No incremental ML diagnostic value demonstrated over candidate rules under these assumptions.

- Prediction** Investigate weak-fault errors, available features and regularisation using training-lot validation.
- Decision** Qualify the real SOP, useful alternative paths, examination scope and continuation costs.
- Evidence** Replace operational assumptions and collect representative complete audits plus fresh prospective evidence.

Retaining rules remains valid. Any shadow study or controlled pilot needs qualification. No policy is selected for operational release.

Source: Benchmark limitations and next evidence; client brief audit requirement. Improvement experiments are proposed.

Presenter notes

Interpretation and next experiments

Time: 9:00–10:00

Improving the 0.497 macro AP alone would not establish business benefit. Investigate errors for cracks, voids and delamination, then test limited feature or regularisation changes using training-lot validation. Avoid repeated optimisation against this already-inspected test cohort. In parallel, engineers must identify where qualified alternative procedures could change completion cost. Lab operations and finance should validate the assumed procedure costs. Quality must confirm report scope, repeat rules and closure. Roughly forty-five stacks per quarter would be selected for full-battery audits, with fewer potentially yielding complete supported labels. Rare-fault evidence will be limited, so pool history carefully and retain a fresh prospective evaluation. Any subsequent shadow study and pilot need qualification first. The experiment supports honest uncertainty and a continued rules-based option, rather than an automatic model release.

Evidence: Benchmark limitations and next evidence; client brief; SYN-OPS-001 section 6.